

Batch 2023

Bachelor of Technology — Computer Science and Engineering

Table 1: Computer Science and Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	38
	Discipline Electives	28
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Computer Science and Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	CS-212	Design of Algorithms	3-0-2-4	4
3	CS-213	Reverse Engineering	0-0-2-1	1
4	CS-214	Computer Organization	3-0-2-4	4
5	CS-304	Formal Language and Automata Theory	3-0-0-3	3
6	CS-309	Information Systems and Databases	3-0-2-4	4
7	CS-305	Artificial Intelligence	3-0-0-3	3
8	CS-312	Operating Systems	3-0-2-4	4
9	CS-313	Computer Networks	3-0-2-4	4
10	CS-302	Paradigms of Programming	3-0-2-4	4
11	CS-303	Software Engineering	2-0-2-3	3
Total				38

Table 3: B. Tech Computer Science and Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Computer Science and Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
9	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Computer Science and Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	IC Basket (choose 1)	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4		CS-214	Computer Organization	3-0-2-4	
5	DC	CS-212	Design of Algorithms	3-0-2-4	4
6	DC	CS-213	Reverse Engineering	0-0-2-1	1
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Computer Science and Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
2	DC	CS-309	Information Systems and Databases	3-0-2-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Computer Science and Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-305	Artificial Intelligence	3-0-0-3	3
2	DC	CS-312	Operating Systems	3-0-2-4	4
3	DC	CS-313	Computer Networks	3-0-2-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Computer Science and Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-302	Paradigms of Programming	3-0-2-4	4
2	DC	CS-303	Software Engineering	2-0-2-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Computer Science and Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Computer Science and Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CS-010	Internship		2
2	MTP	CS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 28 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Computer Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-501	Access Networks	3
2	CS-502	Compiler Design	4
3	CS-506	Cognitive Modeling	3
4	CS-507	Computer Architecture	4
5	CS-508	Introduction to Network Security	3
6	CS-510	Randomized Algorithms	4
7	CS-511	Applied Probability	2
8	CS-512	Matrix Theory	2
9	CS-513	Discrete Mathematics	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	CS-514	Advance Data Structure and Algorithm (CS-580)	4
11	CS-515	Advanced Computer Science Practicum	3
12	CS-516	Exploratory Project	3
13	CS-520	Introduction to Quantum Computing	3
14	CS-521	Introduction to Post Quantum Security	3
15	CS-522	Distributed Algorithms	3
16	CS-524	Computational Complexity Theory	4
17	CS-525	Computational Foundations of Cryptography	3
18	CS-526	Mathematical Foundation of Cryptography	3
19	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
20	CS-541	IoT systems and the Cloud	3
21	CS-542	Design Patterns for Scalable Systems	3
22	CS-544	Formal Concept Analysis: Theory and Practice	3
23	CS-545	Software Design Patterns	3
24	CS-546	Design of Concurrent Software	3
25	CS-547	Network Management Systems	3
26	CS-548	Cloud Networking	3
27	CS-549	Performance Analysis of Computer Networks	3
28	CS-550	Computer Graphics and Geometric Design	3
29	CS-551	Human Computer Interaction	3
30	CS-560	Text Retrieval and Mining	3
31	CS-561	Map Reduce and Big Data	3
32	CS-563	Scalable Data Science	4
33	CS-570	Algorithms for Curves and Surface	3
34	CS-571	Programming Practicum	3
35	CS-591	Special Topics in Knowledge Discovery and Data	1
36	CS-592	Selected Topics in Artificial Intelligence	2
37	CS-601	Probability and Random Processes	4
38	CS-603	Managerial Decision Making	3
39	CS-606	Cognitive Modeling	3
40	CS-609	Speech processing	3
41	CS-611	Program Analysis	4
42	CS-621P	Post Graduate Project-1	12
43	CS-630	Speech Technology	4
44	CS-660	Data Mining for Decision Making	4
45	CS-661	Knowledge Representation and Reasoning	3
46	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
47	CS-670	Kernal Methods for Pattern Recognition	4
48	CS-671	Deep Learning and Applications	4
49	CS-672	Advanced Topics in Deep Learning	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-673	Advanced Computer Vision	4
51	CS-677	Soft Computing	3
52	CS-683	Generative AI	4
53	CS-685	Natural Language Processing	3
54	CS-686	Large Language Models: Core Concepts to Custom	4
55	CS-693	Compressed Sensing and its applications	3
56	DS-201	Data handling and visualization	3
57	DS-301	Mathematical Foundations of Data Science	4
58	DS-314	Probability Theory and Statistics	4
59	DS-401	Optimization for data science	3
60	DS-402	Matrix Computations for Data Science	3
61	DS-403	Introduction to Statistical Learning	3
62	DS-404	Information Security and Privacy	3
63	DS-412	Matrix Computations for Data Science	4
64	DS-415	Optimization for Machine Learning	4
65	DS-416	Fundamentals of Machine Learning (CS-669)	4
66	EE-511	Computer Vision	4
67	EE-608	Digital Image Processing	4
68	MA-221	Numerical Analysis	4
69	MA-222	Applied Linear Programming	4
70	MA-322	Applied Graph Theory	4
71	MA-546	Introduction to Quantitative Finance	4
72	MA-552	Number Theory	3
73	MA-605	Statistical Data Analysis	3
<i>Compulsory project components — counted as DE, not selectable</i>			
74	CS-498P	Major Technical Project - I (compulsory; counted as DE)	4
75	CS-499P	Major Technical Project - II (compulsory; counted as DE)	4